AP CALCULUS AB – UNIT 4

4.6 L'Hopital's Rule

Name: ____________ **Class:** ____________ **Date:** ____________

Lesson Overview and Learning Objectives

- Recognize the indeterminate quotient forms $0/0$ and ∞/∞.
- Verify the hypotheses of L'Hopital's Rule before differentiating numerator and denominator.
- Evaluate indeterminate limits, including repeated applications when justified.
- Rewrite selected products, differences, and powers into allowed quotient forms.

Keys: Essential Knowledge and Vocabulary

Allowed forms	Direct substitution must give $0/0$ or ∞/∞.
L'Hopital's Rule	If the hypotheses hold, $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ when the derivative ratio limit exists or is infinite.
Not a quotient rule	Differentiate numerator and denominator separately; do not differentiate the quotient by the Quotient Rule.
Repeated use	Recheck the form after each application before applying the rule again.
Other forms	Products $0 \cdot \infty$, differences $\infty - \infty$, and powers require algebraic or logarithmic transformation first.
Alternative methods	Factoring, rationalizing, known limits, or direct substitution may be simpler and should be considered first.

Explanation, Strategy, and Common Errors

- Always show the substitution that produces an allowed indeterminate form.
- Differentiate the numerator and denominator as separate functions and then reevaluate the resulting limit.
- A form such as $1/0$, $0/\infty$, or $\infty/0$ is not indeterminate and does not justify the rule.
- For 1^∞, 0^0, or ∞^0, take logarithms, evaluate the logarithmic product, and exponentiate at the end.
- State when repeated application is justified rather than applying derivatives mechanically.

Solved Example: One application

Evaluate $\lim_{x\to 0} \frac{e^{2x}-1}{x}$.

Answer and Reasoning

The form is $0/0$. Differentiate top and bottom: $\lim 2e^{2x} = 2$.

Solved Example: Repeated application

Evaluate $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

Answer and Reasoning

It is $0/0$. First application gives $\lim \sin x/(2x)$, still $0/0$. A second gives $\lim \cos x/2 = 1/2$.

Solved Example: Infinity over infinity

Evaluate $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$.

Answer and Reasoning

The form is ∞/∞. The derivative ratio is $(1/x)/(1/(2\sqrt{x})) = 2/\sqrt{x} \to 0$.

Solved Example: Transform a product

Evaluate $\lim_{x\to 0^+} x \ln x$.

Answer and Reasoning

Rewrite as $\ln x/(1/x)$, an $-\infty/\infty$ form. The derivative ratio is $(1/x)/(-1/x^2) = -x \to 0$.

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For LH́opital problems, verify an allowed indeterminate form first.

Leveled Practice

Easy Foundations

Build fluency with the definition, notation, and one-step applications.

1. Identify the form of $\lim_{x\to 0} \frac{\sin x}{x}$.

2. Evaluate $\lim_{x\to 0} \frac{e^x - 1}{x}$.

3. Evaluate $\lim_{x\to\infty} \frac{x}{e^x}$.

4. Explain why $\lim_{x\to 2} \frac{x+1}{x-2}$ does not permit the rule.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Evaluate $\lim_{x\to 0} \frac{\ln(1+x)}{x}$.

6. Evaluate $\lim_{x\to 0} \frac{x-\sin x}{x^3}$.

7. Evaluate $\lim_{x\to \infty} \frac{x^3}{e^x}$.

8. Evaluate $\lim_{x\to \infty} \frac{\ln x}{x}$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Evaluate $\lim_{x\to 0} \frac{\tan x - x}{x^3}$.

10. Evaluate $\lim_{x\to\infty} x(e^{1/x} - 1)$.

11. Evaluate $\lim_{x\to 0^+} x^x$.

12. Evaluate $\lim_{x\to\infty} \left(1 + \frac{3}{x}\right)^x$.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Evaluate $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$.

14. Evaluate $\lim_{x\to\infty} x\left(\frac{\pi}{2} - \arctan x\right)$.

15. Evaluate $\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Suppose $f(0)=g(0)=0$, $g'(0)\neq 0$, and both are differentiable near 0. State a condition guaranteeing $\lim f/g = f'(0)/g'(0)$.

17. Evaluate $\lim_{x\to 0^+}(\sin x)^x$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. L'Hopital's Rule may be applied directly when substitution gives

(A) $3/0$
(B) $0/5$
(C) $0/0$
(D) $\infty - \infty$

2. $\lim_{x\to 0}(e^{4x} - 1)/x$ equals

(A) 1
(B) 4
(C) e^4
(D) 0

3. After one application, a limit is still $0/0$. The correct next step is

(A) declare DNE
(B) apply the rule again after verifying hypotheses
(C) use Quotient Rule
(D) replace the limit by zero

4. To evaluate $x \ln x$ as $x \to 0^+$, a useful rewrite is

(A) $x/\ln x$
(B) $\ln x/(1/x)$
(C) $\ln(x/x)$
(D) $1/(x \ln x)$

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. $\lim_{x\to 0} \frac{\sin(2x)-2x}{x^3}$ is

(A) $-4/3$
(B) $-2/3$
(C) 0
(D) $4/3$

6. $\lim_{x\to\infty} \frac{x^5}{e^{0.2x}}$ is

(A) 0
(B) 1
(C) 5
(D) ∞

AP-Style Free-Response Practice

No Calculator

Consider $L = \lim_{x\to 0} \frac{e^{2x}-1-2x}{x^2}$.

(a) Verify the initial indeterminate form. [1 points]

(b) Use L'Hopital's Rule to evaluate L. [3 points]

(c) Explain why two applications are justified. [2 points]

Calculator Permitted

A calculator is used to investigate $K = \lim\limits_{x\to\infty} x\left(e^{3/x} - 1\right)$.

(a) Rewrite K as an indeterminate quotient. [2 points]

(b) Evaluate the limit. [2 points]

(c) Give numerical evidence using a large value such as $x = 10^4$. [2 points]

Original instructional resource prepared for Education City High School. Not an official College Board publication.